

Assertion and Reasoning Questions – World of Numbers (Grade 9)

Instructions:

For each question, choose the correct option:

- A. Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A.
 - B. Both A and R are true, but R is not the correct explanation of A.
 - C. A is true, but R is false.
 - D. A is false, but R is true.
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1.

Assertion (A): Every natural number is a whole number.

Reason (R): Whole numbers include 0 and all natural numbers.

2.

Assertion (A): The decimal expansion of $\frac{1}{8}$ is terminating.

Reason (R): The denominator 8 can be expressed as 2^3 .

3.

Assertion (A): $\sqrt{25}$ is a rational number.

Reason (R): The square root of every perfect square is an integer.

4.

Assertion (A): The sum of two irrational numbers is always irrational.

Reason (R): Irrational numbers cannot be expressed as $\frac{p}{q}$, where $q \neq 0$.

5.

Assertion (A): Zero is a rational number.

Reason (R): Zero can be written as $\frac{0}{1}$.

6.

Assertion (A): Every integer is a rational number.

Reason (R): Any integer n can be written in the form $\frac{n}{1}$.

7.

Assertion (A): The decimal expansion of $\frac{7}{15}$ is non-terminating recurring.

Reason (R): The denominator 15 contains a prime factor other than 2 or 5.

8.

Assertion (A): $\sqrt{2}$ is an irrational number.

Reason (R): $\sqrt{2}$ cannot be expressed in the form p/q , where p and q are integers and $q \neq 0$.

9.

Assertion (A): The product of two rational numbers is always a rational number.

Reason (R): Rational numbers are closed under multiplication.

10.

Assertion (A): Every whole number is a natural number.

Reason (R): The set of whole numbers contains 0.